Of Motherboards and the military...

In a freewheeling chat over coffee with Jack Cheng, we got the lowdown on everything from the latest Z68 chipset to how motherboards are actually made

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d Tell us a bit about Z68; improvements, what's different in your implementations?
In our Z68 series the major highlight of course is based on special features from Intel; whether it's swappable graphics, or Smart Response technology. For our Z68 boards we also took some of the best things from P67 as well. For example the ASUS UEFI bios is a little better than competitors. Our competition has simply put just an application and claims it to be a BIOS. Who is going to use this? After you boot into Windows why would you change these settings? And even if you do, you will require a restart, so you see the irony there. With these hybrid EFI implementations you can only install OS from DVD drive connected to the native SATA port. USB flash installation is a no go. USB optical drive will fail too.

Another new thing is smart response. In order to use smart response you have to have spindle hard drive and SSD of course. Many reviewers tend to use any hard drive and any SSD, and complain about no performance improvement. Being a pure software implementation, it's necessary for the performance of the SSD to be at least double of the spindle HDD.

Switchable graphics is the next area. Normally while encoding your discrete card will be at 100 percent load. But by using integrated graphics it will create 10% of the CPU load and no load on the graphic card. Rendering wise I have to say, Intel rendering is good but it's no match with high end NVIDIA card. But with internal graphics you save power and you can actually do encoding while you play Warcraft.

Jack Cheng, is a member of the ASUS HQ motherboard team. He is a motherboard expert and the writer of all ASUS motherboard media reviewer guides.

I just outlined the difference between d-Mode and i-Mode. So on our high end boards we don't provide the i-Mode at all. For two reasons. One is we don't see any need and two, because they have way too many features, we cannot fit any more. I personally use d-Mode. The only purpose I see of i-Mode is I can disable lucid and run 6 monitors while enable lucid any-time and play my games on one monitor.

d Here's a scenario. Suppose you're rendering something and you're using i-Graphics to render but using the d-Port, is there a performance drop?
There is not going to be a performance drop in terms of GPU. Even if there is, it'll be very minor. All the integrated pipelines are inside the CPU, you will see some impact on CPU performanc as CPU packets will have to wait.

d Of course no one would want to do it, but out of curiosity what if you do the inverse? Have the Discreet card do the rendering or encoding and push it through the i port?
I've never tried it (laughs)

d What are the various steps involved in bringing a motherboard to market.
Intel has a roadmap of when a chipset is going to be announced and when it's going to be available to motherboard manufacturers. Even before we get a reference board we have to think about things like, "Intel is giving us this, what can we add on from there?". We have an innovation team who thinks of what to add on, and a team that decides whether those innovations have actual applications. If there are no applications, they are rejected. We also keep an eye out for reports from reviewers and users about features from previous generation boards.

d What about competition?
Of course. We study every generation from competitors, but rather than copy, we try to improve upon the good things.

d How drastic a performance difference can a user experience, if he goes for a ROG board as opposed to a standard board with overclocking features?
Without overclocking everything is going to be the same. Same chipset, same everything. However, over-clocking is where the difference comes across. Generally speaking with the best components you can get a 10% performance boost on overclocking with Sandybridge. And stability for gaming. Our tough series goes through temperature tests, stress tests, while gaming boards go through torture tests!

d Speaking of which when mother board manufacturers say "military" grade is it a marketing term ?
No our military components are actually used by the US army. It's not a marketing term. These components have to go through even more rigorous tests chamber tests like electric shock, vibration tests. Entire boards have to endure the tests not just parts on the board individually. Others are just claiming to be military only recently. d